From Iss to I-distance: Quantifying DNA Evolution with Information Theory

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Abstract. The DNA evolutionary process is traditionally inferred by evaluating certain substitution models for homologous sequences, which is often computationally intensive and black-box. Here, we report that the Information-based index of substitution saturation (Iss), conventionally used as a mere index for substitution saturation diagnosis, in fact is a universal informational-theoretic ruler for genetic divergence. Through extensive evolutionary simulations, we demonstrate that Iss exhibits a precise, model-dependent relationship with p-distance, effectively decoding the hidden evolutionary signature embedded within sequence alignments. Consequently, we repurpose Iss from a saturation watchdog into a powerful and model-agnostic statistics and purpose for a new name, informational distance (abbr. i-distance). Furthermore, we show that a combined use of i-distance and traditional statistics can directly distinguish evolutionary models (e.g., JC69, K2P, GTR), diagnose topology biases, and even signal the presence of mixed evolutionary processes. This discovery bridges information theory, statistics, and molecular phylogenetics, laying the foundation for a new paradigm, Informational Molecular Evolution, that allows for the visual and quantitative dissection of evolutionary forces directly from sequence data.

Keywords. Molecular phylogenetics, molecular evolution, information theory, informational distance, evolutionary simulation, substitution saturation.

Introduction

Unraveling the history and mechanisms of molecular evolution relies on comparing homologous sequence alignments. In the past 60 years' development of computational molecular evolution, the only path of this effort is the selection of an appropriate substitution model, which is inherently a statistical model-fitting problem. Current approaches, based on likelihood maximization and information criteria, or the Markov Chain Monte Carlo, while powerful, operate as "black boxes". While outputting a best-fit model, they offer limited intuitive insight into *why* that model best fits or how the evolutionary process *visibly* manifests in the data. Moreover, the analysis of saturated sequences remains a big challenge, as standard genetic distance estimates become indeterminate or completely invalid facing saturated alignments.

The Index of substitution saturation (Iss) statistic, introduced by Xia et al.[1, 2] in 2003, was designed to address the latter problem. It measures the degree of saturation by comparing the average informational entropy of observed substitutions (Hm) to the maximum possible entropy of the observed base (Hfss). For over two decades, its utility has only been confined to a simple decision: whether sequences are too saturated for reliable phylogenetic inference. The critical value for this decision (Iss.c) is derived from complex simulations, making the function hard to reproduce and underlying richness of Iss overlooked.

In this work, we demonstrate that this conventional view represents a profound underutilization of the Iss statistic. We show that Iss is not merely a gatekeeper but a versatile informational-theoretic descriptor of sequence polymorphism relevant to multiple variables of the substitution process. This redefinition unlocks an entirely new suite of its applications, transforming Iss into a powerful and intuitive tool for diagnosing multiple perspectives of the very evolutionary processes that traditional methods seek to model.

Results

1. Establishing the Fundamental Behavior of Iss

We began by systematically replicating and extending the original framework for calculating Iss.c. Large-scale simulations were automatically completed under various models (JC69, K2P, F81, F84, HKY85, TN93, GTR), model rate and base frequency parameters, gamma distribution parameters, tree topologies, and branch lengths, with their Iss values automatically calculated. We noticed that Iss can be influenced by all variables in our simulations, and monotonically increases to tree lengths (Figure 1), and an amazing pattern emerged: Under same tree topologies and other model parameters, the median of Iss distribution is irrelevant to sequence lengths, but increases with the growth of tree lengths, while its deviation narrows with the increase of sequence lengths (Figure 2A). What is even better, across all models, Iss formed a smooth, continuous curve when plotted against the mean p-distance of the 4-sequence set (Figure 2B). This robust, monotonic relationship revealed that Iss is a precise proxy for genetic divergence, a role far exceeding its original binary saturation-detection purpose.

When in 2-sequence case, each site's entropy is restricted to zero (same bases) or one (divergent bases), indicating the degeneration of Hm to the common pairwise p-distance (Figure 3B), while the base-frequency information included in Hfss makes the scatter points of Iss rising a bit when base frequencies are unequal (Figure 3A), despite its general strong correlation to pairwise p-distance. This is also observed in our simulation.

It is worth noting that by definition, Iss is lower than one in an ideal situation, but its systematic variation, as deducted by Xia's work, may let it be beyond this limit when being observed.

2. Unveiling the Dual Nature of Iss: An Informational-Theoretic Measure of DNA polymorphism

We discovered that the point of maximum saturation (Iss = 1) corresponds precisely to the theoretical maximum p-distance under a given model (e.g., ~0.75 for JC69), which can also be deducted with the Markov chain steady states of base frequencies. This critical finding bridges the informational-theoretic definition of Iss with the statistical concept of p-distance and the steady-state of nucleotide substitution Markov processes. It reveals that Iss encapsulates the same core information—sequence polymorphism—as p-distance, but through a model-latent and informational-theoretic lens. This dual nature explains its fundamental behavior and provides a theoretical anchor for its application.

3. Decoding the Hidden Layer: Iss as a Reservoir of Evolutionary Information and Model-Specific Signatures

Having established Iss's model-decoding capacity, we now redefine it as Informational distance

(i-distance), a term better fitting its role as an evolutionary divergence measure. We hypothesized that if i-distance captures divergence, its relationship with p-distance might be shaped by the underlying evolutionary model. We decomposed the p-distance into its constituent components (e.g., transition vs. transversion p-distance for K2P; all six substitution-type distances for more complex models, e.g. GTR). Plotting i-distance against these components revealed model-specific fingerprints: nearly identical trajectories for JC69, two distinct clusters for K2P, and six unique trajectories for an unconstrained GTR model (Figure 4).

Furthermore, Xia's work revealed that Iss.c is sensitive to tree topology, with asymmetric topologies having lower Iss.c value than symmetric ones, indicating that Iss itself may be impacted by tree topology. Our work further proved that asymmetric trees yield systematically higher Iss values than symmetric trees of equal total length, revealing a previously hidden signature of evolutionary history.

Discussion

We have demonstrated that the Iss statistic is a rich, information-theoretic measure of evolutionary divergence, not a simple threshold. This redefinition transforms it from a simple tool into a core diagnostic for molecular evolution with great empirical potential. The ability to visually distinguish between evolutionary models and to detect heterogeneity directly from Iss-p plots indicates a paradigm shift from only black-box model-fitting to its collaboration with transparent model-diagnosis. Also, we propose renaming the Iss statistic as 'Informational distance' (i-distance) to better reflect its role as an information-theoretical measure of DNA polymorphism and sequence divergence, corresponding to proportional distance (p-distance).

This work bridges the fields of information theory, statistics, and evolutionary biology. The Iss statistic acts as a translator, converting the statistical patterns of sequence alignment into informational terms, which we show are exquisitely sensitive to the parameters and processes of evolution.

We propose that these findings establish the foundation for a new sub-discipline: Informational Molecular Evolution. This framework leverages information-theoretic[3] quantities to directly visualize, quantify, and diagnose evolutionary forces. Future applications are vast, including illuminating topological biases, resolving recombination and hybrids, analyzing saturated sequences, and developing information-based distance metrics immune to overflow. By looking at sequences through this new perspective, we gain an intuitive, powerful, and fundamentally different way to understand the molecular evolutionary process.

We argue that this new paradigm must be bipedal, supported by its left and right arms: the basic statistics of information theory and the generative power of evolutionary simulation. In an era of abundant computing power, the path forward can be fully empirical: simulate under hypothesis, and compare the informational statistics of simulated outcomes to real data. This is the core of a powerful new hypothesis-testing framework. By looking at sequences through this informational lens, we gain an intuitive and profoundly different way to understand molecular evolution.

Methods

Large-scale simulations were performed using Evolver in PAML v4.10.7[4] to generate sequence alignments under multiple models (JC69, K2P, HKY, F81, F84, TN93, GTR[5]), gamma distributions, tree topologies (symmetric or asymmetric), and branch lengths, with the Iss statistic and p-distances calculated according to the definition in Xia et al. in a custom Python pipeline. All analysis code, simulation scripts, and plotting routines are available on Github. The heatmap was generated with matplotlib's imshow function, and other plots in this work were generated with plotnine in Python.

Data Avaliability

All source codes of python scripts are available on the Github depository: https://github.com/Gipsy-The-Sheller/Iss-simulation-scripts.

The TAR packages of all simulation results , vector graphs of individual plots (in SVG format) and partial corresponding bitmaps (in PNG format) are available on Zenodo via DOI: https://doi.org/10.5281/zenodo.17071721.

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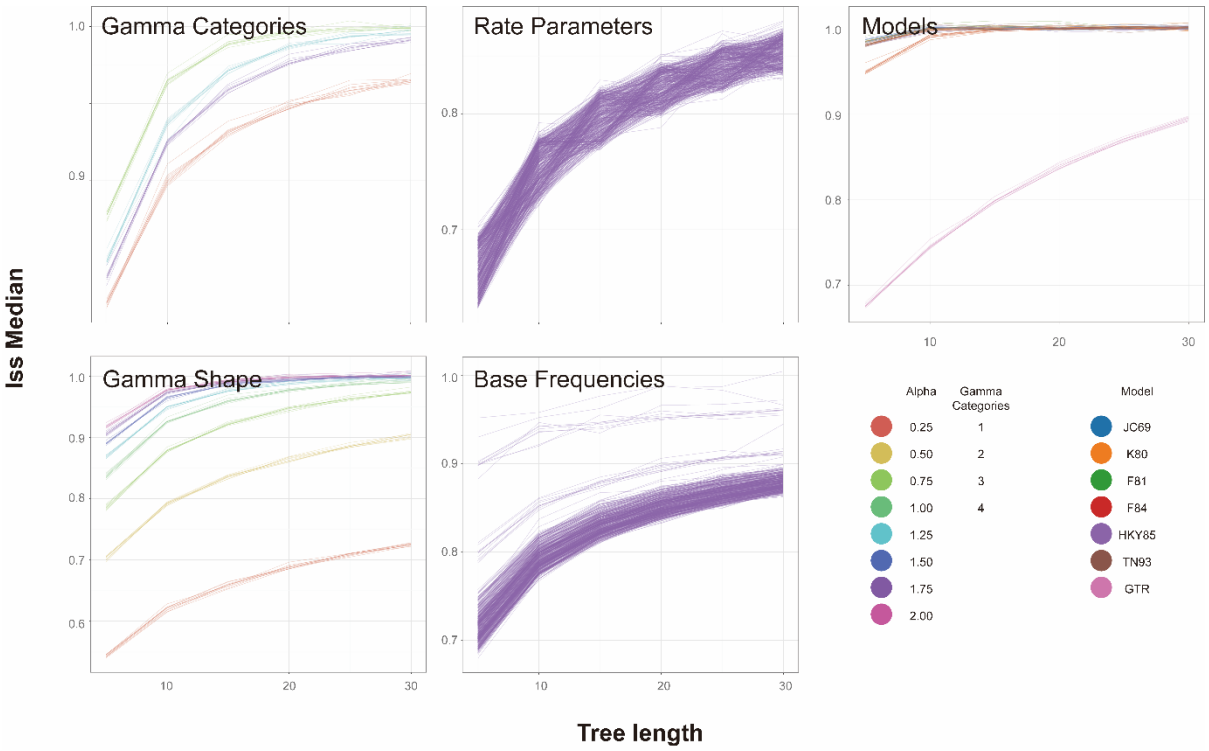


Figure 1: Iss can be impacted by various evolutionary variables.
Line charts were plotted by plotting Iss median values of single simulations versus tree lengths, under a 4-leaf symmetric topology.

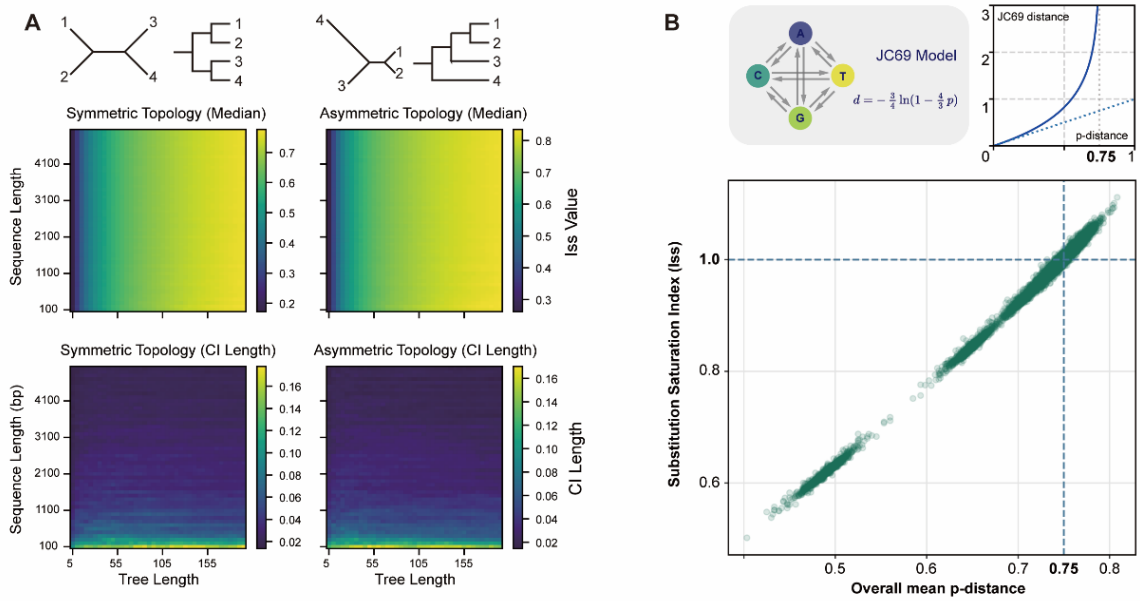


Figure 2: Iss is an information-theoretic measure of DNA polymorphism.

A. Heatmap of Iss values from simulations under HKY85 model and both symmetric and asymmetric topologies, showing a smooth gradient of with increasing sequence length (y-axis) and increasing tree length (x-axis).

B. Scatter plot of Iss versus mean p-distance for sequences evolved under JC69 model. Above is a schematic diagram of JC69 Markov chain and the relationship between JC69 distance and p-distance.

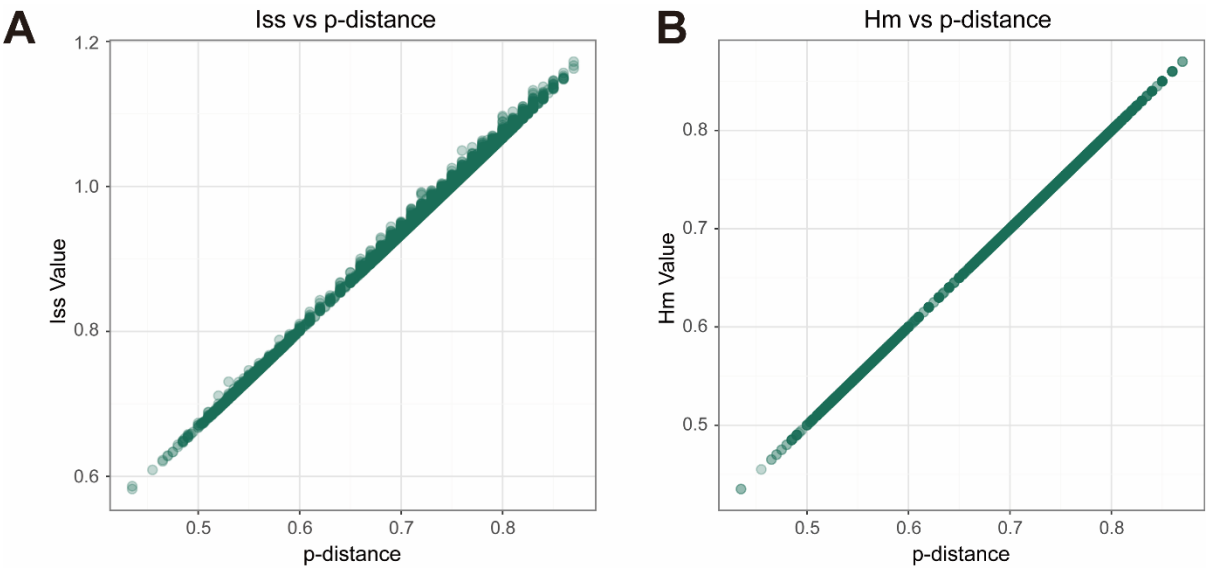


Figure 3: Behavior of Iss and Hm vs p-distance curves when the number of sequences is two under JC69 model.

A. pairwise Iss vs pairwise p-distances.

B. pairwise Hm is equal to pairwise p-distances.

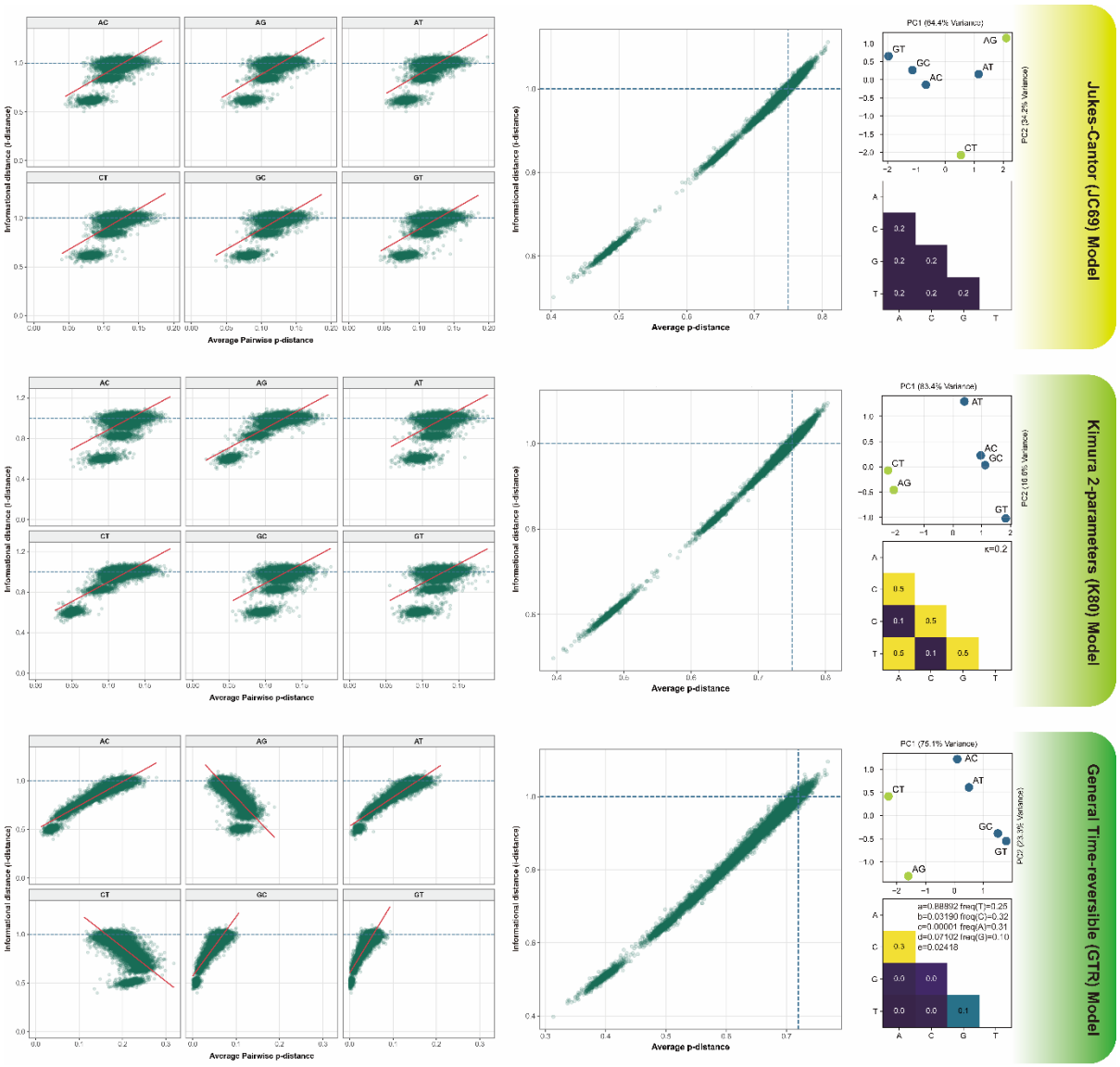


Figure 4: I-distance (informational distance, evolved from Iss) decodes evolutionary model signatures.

left. I-distance vs. 6 p-distances of different substitutions under JC69, K80 and GTR model.

middle. I-distance vs. overall mean p-distances. Vertical auxiliary lines show expected p-distances of fully saturated sequences based on stationary base frequencies of models' Markov chains.

right. PCA plot based on mean I-distance and p-distance values for different substitution types, with transitions marked as blue points and transversions marked as green points, and partial substitution rate matrix of model parameters used in simulations.

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